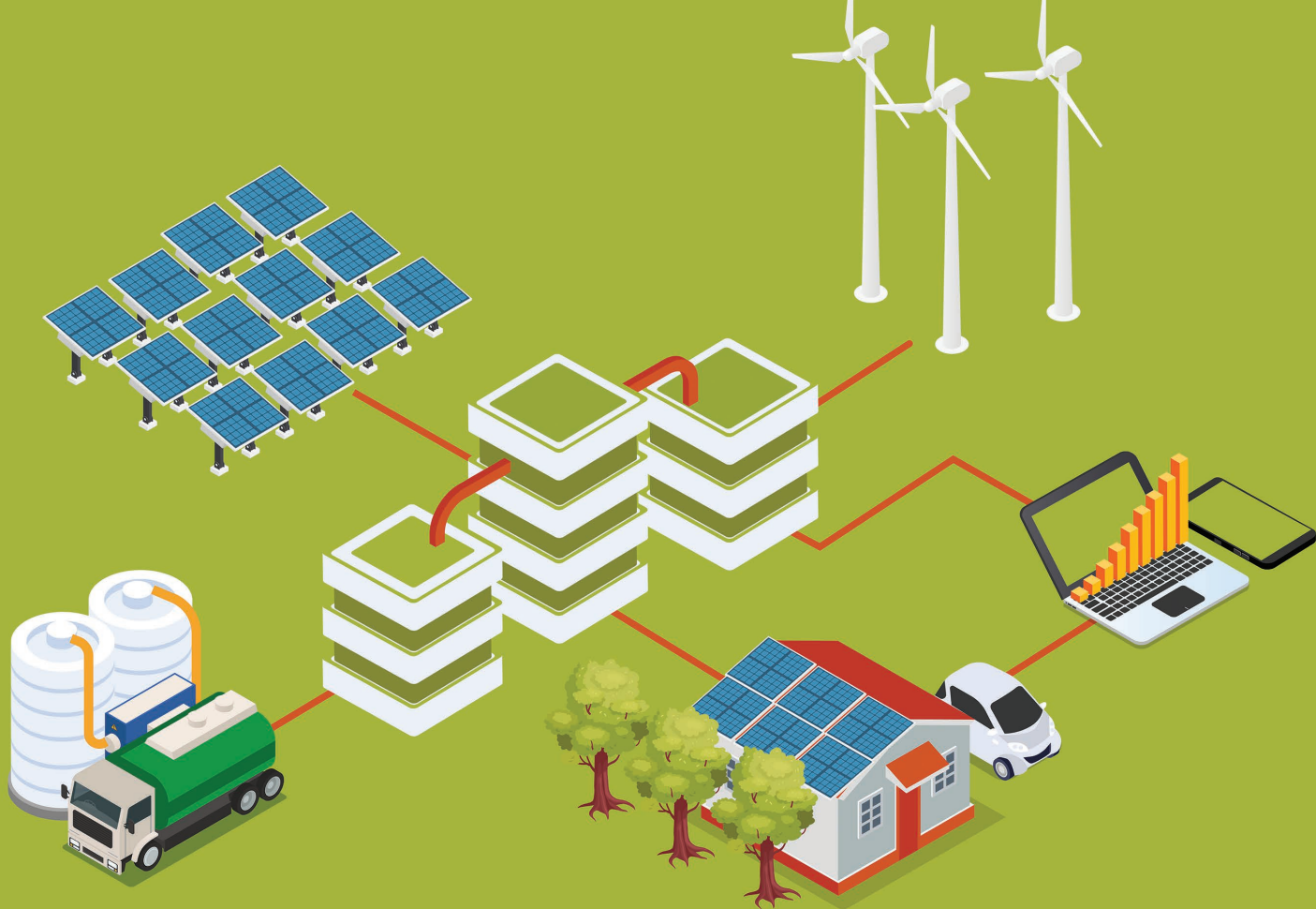




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AI Data Centers and the Grid

New Challenges and Opportunities for Reliability and Resilience

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THE RAPID PROLIFERATION OF ARTIFICIAL intelligence (AI) applications and societal reliance on computing is driving an unprecedented surge in electricity demand, with AI data centers emerging as one of the fastest-growing and most energy-intensive load categories. These facilities often draw hundreds or even thousands of megawatts (MW) of power and exhibit demand characteristics that differ markedly from traditional large loads such as aluminum smelters.

Introduction

Unlike traditional industrial facilities, AI data centers concentrate synchronized power demands over relatively short time frames, often in regions with constrained transmission capacity or limited generation headroom. Their unprecedented scale and variability are forcing a reexamination of long-standing approaches to system planning, operation, and regulatory oversight.

Compounding the challenges introduced by AI data centers, the grid itself is facing concurrent transformative pressures, including the integration of power electronically coupled resources to complement conventional synchronous generation, the deployment of flexible grid assets, such as battery energy storage (BES), to manage variability and provide ancillary services, and the expansion of grid measurement and monitoring systems to improve situational awareness, operational decision making, and control.

The combined effect of these pressures is reshaping the bulk power system (BPS), creating conditions in which large concentrated loads exert an outsized influence. The introduction of AI data centers acts as a catalyst for these trends, accelerating demand growth across communities and regions. These trends also intersect with growing planning and operational challenges, including

- addressing bottlenecks in expanding large-scale transmission infrastructure, such as supply chain constraints, land availability, and regulatory hurdles, that limit the ability to meet growing demand
- managing resource adequacy risks from uncertainty in generation and load across multiple timescales
- mitigating evolving cyber and physical security threats that require ongoing oversight
- adapting to extreme weather events, including wildfires, hurricanes, and flooding, that are increasing in frequency and severity.

But the imperative remains: the electricity sector must supply end-use customers with

reliable, resilient, and affordable electricity while simultaneously supporting the advancement of modern society.

The U.S. National AI Action Plan outlines an urgent strategy to align the nation's electric grid with the growing energy requirements of AI technologies.¹ It emphasizes that the grid must keep pace with rising demand from AI data centers, which are central to sustaining innovation and U.S. leadership in the field. To achieve this, the plan prioritizes upgrading transmission infrastructure, enhancing grid resilience, and deploying advanced dispatchable generation, such as small modular reactors. Translating these priorities into action will hinge on targeted investment and workforce development. Without enough skilled power system engineers, even well-planned upgrades to transmission, generation, and grid resilience cannot be implemented at the scale required.

Accommodating the rapid growth of AI loads requires balancing two critical objectives: supporting the expansion of the AI economy while ensuring grid reliability and resilience. Developing strategies that incorporate advanced planning, flexible operations, and coordinated stakeholder engagement will be essential to maintaining this balance. The success of these strategies will determine the grid's ability to support a future increasingly shaped by AI.

AI Data Centers Driving Electricity Demand Growth

For nearly two decades, electricity demand in the United States remained relatively flat. From the early 2000s through the mid-2010s, national consumption growth slowed to near-zero levels. This plateau reflected energy efficiency improvements, structural economic changes, and the widespread replacement of inefficient technologies. Advances in data center power utilization largely offset rising demand for computation, thanks to more efficient server hardware, system architectures, and cooling technologies. During this period, the data center landscape included a mix of cloud computing, enterprise, and emerging blockchain/crypto operations, each with distinct load profiles that collectively shaped electricity demand. Between 2014 and 2016, U.S. data center electricity use hovered around 60 TWh annually, despite rapid digital transformation. In response to this stagnation, utilities adopted more conservative planning practices, regulators

¹<https://www.ai.gov/action-plan>.

recalibrated standards, and investment remained subdued. Consequently, transmission expansion slowed, and modest growth in generation capacity caused reserve margins to tighten.

This equilibrium is unraveling. Since 2021, utilities and grid operators have sharply revised load forecasts upward, signaling a new era of electricity demand growth. While electrification in industry, transportation, and hydrogen production contributes to this shift, the most immediate driver is the expansion of data centers. The number of operational data center facilities worldwide rose from approximately 8,000 in 2021 to nearly 12,000 by early 2025. The United States hosts the largest share, with more than 5,400 sites.

Data center electricity consumption in the United States tripled from 60 TWh in 2017 to 176 TWh in 2023, accounting for roughly 4.4% of total usage, as shown in Figure 1. According to the Lawrence Berkeley National Laboratory, forecasts suggest that this level could more than triple again by 2028, reaching 580 TWh, or about 12% of total U.S. electricity consumption. AI, with its GPU-accelerated training and inference, imposes far greater demands for power, cooling, and throughput than conventional IT, making it the primary driver of this growth. For context, a single generative-AI query can consume nearly 10 times the electricity of a standard web search.

The rapid pace of data center deployment creates acute planning challenges. Hyperscale

AI data centers, with peak loads of 100 MW or more, can be built and commissioned within one to two years. But the transmission, generation, and substation infrastructure needed to support them often lags behind. New interconnection requests routinely span 100–1,000 MW, roughly equivalent to the power needs of a small city. These requests are often concentrated in specific regions and appear with limited notice. Unlike gradual residential or commercial growth, these surges stress the grid and complicate resource adequacy. More importantly, the technical characteristics of AI loads—power density, variability, synchronization, and potential resonance effects—pose system reliability challenges distinct from conventional computing infrastructure. These challenges are forcing utilities, regulators, and system operators to reconsider long-standing planning and operational practices.

AI Infrastructure Load Characteristics

Specialized computing units, known as *AI accelerators*, lie at the heart of AI data centers. Most of these accelerators are based on graphics processing units (GPUs), purpose-built for the massively parallel computations that AI workloads require. Racks of GPU-powered accelerators are supported by extensive cooling infrastructure and robust power delivery systems, including uninterruptible power supplies (UPSs). Networking, lighting, and

site management systems are also present, but their electricity demand is generally small compared with compute and cooling. Cooling equipment, including chillers, liquid loops, and air-handling units, also consumes substantial amounts of power, highlighting why these facilities are such large and dynamic electricity consumers.

Modern AI accelerators can spike their electricity use almost instantaneously. At scale, this creates significant challenges for the grid; repeated surges can strain equipment, complicate balancing, and even risk triggering system-wide oscillations. In AI data centers, accelerators account for most of the

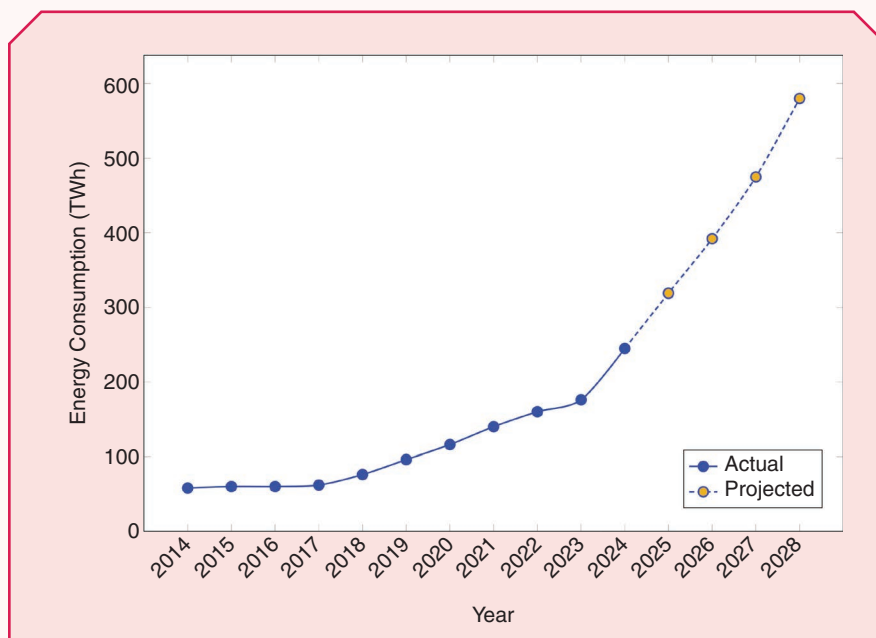


figure 1. The actual and forecasted growth in U.S. data center electricity consumption, 2014–2028.

electricity use, running training jobs that cycle between intense bursts of computation and brief pauses for synchronization.

During computation, AI accelerators draw power near their maximum thermal limits, with demand dropping sharply during the synchronization pauses. These cycles occur over intervals ranging from fractions of a second to tens of seconds. They produce large and highly periodic power swings that can reach tens or even hundreds of megawatts per facility. Critically, these power swings fall in the 0.2–3-Hz range, overlapping with the grid's natural oscillation frequencies, arising from the electromechanical dynamics of synchronous generators. This means that large periodic data center loads could unintentionally excite system-wide oscillations, putting grid reliability at risk. [Figure 2\(a\)](#) illustrates a typical AI power demand profile in the time domain, which is markedly different from conventional information technology workloads, where consumption is relatively steady. [Figure 2\(b\)](#) shows the frequency-domain representation of GPU power consumption during AI training, illustrating spectral content in the relevant frequency range.

Hence, AI data centers aren't just big consumers; they can introduce unexpected stresses to the grid, requiring operators to manage their fluctuations in real time. Left unchecked, their dynamic demand characteristics pose risks to BPS reliability and quality of service. However, careful management of on-site resources within these facilities, such as batteries, could create new opportunities for supporting grid operations and control. There is growing interest in the idea that data centers could act as flexible power system resources, adjusting their load in real time to help balance supply and maintain

grid stability. Harnessing this potential would allow the grid to accommodate rapid AI growth while enhancing overall system reliability, a crucial consideration as these facilities expand worldwide.

Regional Load Growth Patterns From AI Data Center Deployment

Although AI loads are driving national electricity demand, that growth is unevenly distributed. [Figure 3](#) illustrates data center operating capacity by state, organized according to Electric

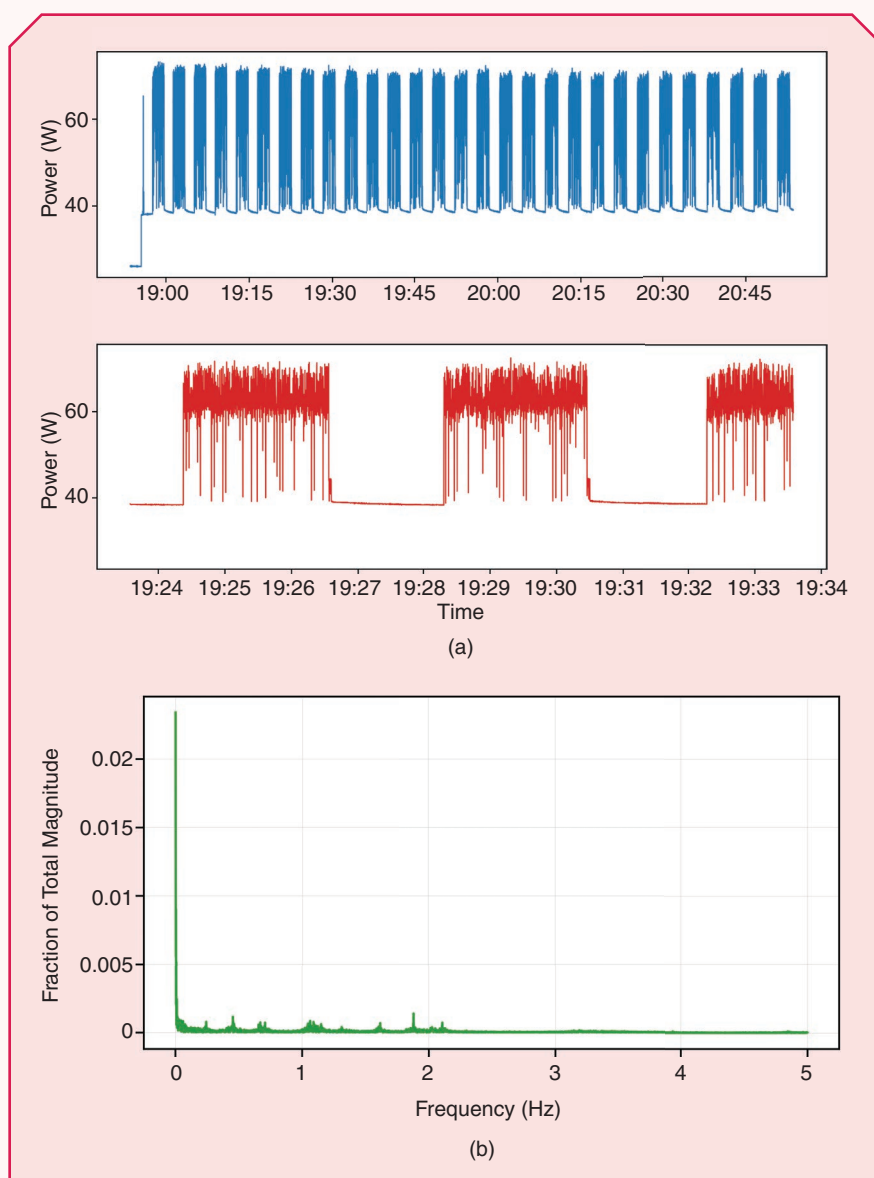


figure 2. (a) An example GPU power consumption profile during an AI training workload. The top panel shows a 2-h segment of GPU power draw, while the bottom panel zooms in on a 10-minute interval to provide a fine-grained view of the characteristic cyclical fluctuations between compute-intensive and synchronization phases. (b) A frequency-domain representation (FFT) of the GPU power consumption during AI training.

Reliability Organization (ERO) regions, with [Table 1](#) summarizing regional totals. Most of the new load is concentrated within existing clusters, particularly in Northern Virginia, which hosts the world's largest concentration of data centers, and in Texas, where recently approved and queued projects represent tens of gigawatts of additional demand.

The expansion of AI data centers across various regions is creating new and complex challenges for long-term electricity planning. Unlike other load categories, AI data centers are deployed rapidly, often clustered in specific areas, and have a limited operational track record. Their technology evolves quickly, and future demand patterns are unclear, leaving grid stakeholders unsure how best to integrate

and serve these loads. Engineers and planners, for example, often have little guidance on how to model and study these facilities accurately across the range of timescales required. Together, these factors are reshaping system planning, operations, and oversight, and they require innovative approaches to ensure grid reliability and resilience moving forward.

Reliability Risks of AI Data Centers and How to Mitigate Them

Continuous and reliable flow of electricity to end consumers is foundational to modern society. Sustaining this foundation requires understanding the grid's performance across multiple dimensions. Power system reliability can be described in terms of

- *adequacy*: the availability of sufficient generation, transmission, and distribution resources to meet demand under steady-state conditions
- *security (operating reliability)*: the ability to maintain stability and operate within limits during normal operating conditions and during defined system disturbances such as faults or loss of generation or load.

Both pillars of reliability are increasingly stressed by extreme weather, generation capacity shortfalls, and the variability of renewables. The integration of large amounts of inverter-based generation and loads is

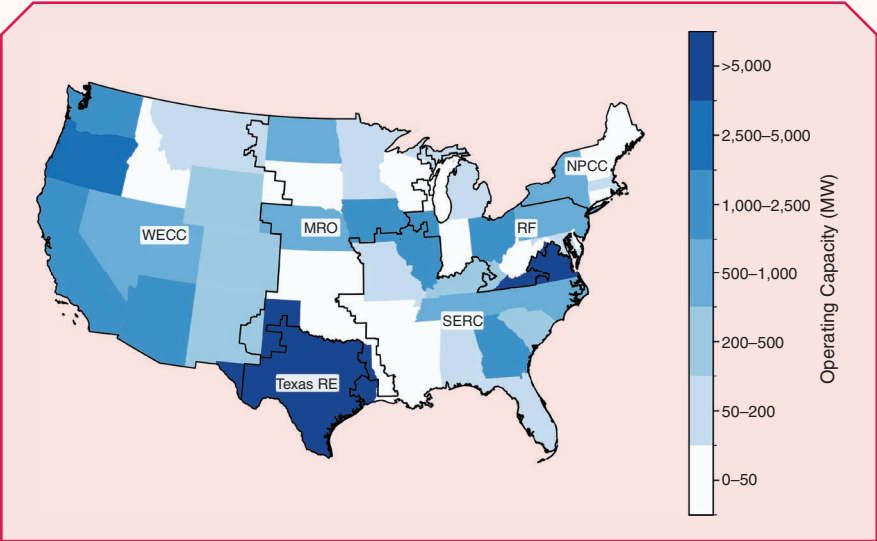


figure 3. Contiguous U.S. data center operating capacity by state with ERO region boundaries. WECC: Western Electricity Coordinating Council; MRO: Midwest Reliability Organization; Texas RE: Texas Reliability Entity; RF: reliability first; NPCC: Northeast Power Coordinating Council; SERC: SERC Reliability Corporation.

table 1 Data center capacity by ERO region.			
ERO Region	Operating (MW)	In Construction (MW)	Planned (MW)*
SERC (Southeast)	14,259	6,881	53,173
WECC (West)	10,915	3,964	17,233
Texas RE (Texas)	8,904	6,237	24,703
RF (Mid-Atlantic)	5,106	4,251	28,754
MRO (Midwest)	3,544	1,752	7,695
NPCC (Northeast)	1,016	199	962

*Planned capacity reflects data center projects listed in regional interconnection queues and publicly reported planning datasets. These values represent interconnection requests rather than committed or fully approved builds and are subject to uncertainty as many planned projects may be delayed, downsized, or not ultimately realized.

changing the fundamental nature of power system design and control, while aging transmission and distribution infrastructure limits system flexibility. The expansion of large loads such as AI data centers adds another layer of complexity atop these enduring stressors.

How AI Data Centers Impact Grid Reliability

To examine how AI data centers affect the BPS, we organize their impacts around the two pillars of reliability. Long-term planning and load forecasting primarily influence adequacy, while operational and real-time challenges are most closely linked to security. Grid stability and power quality concerns affect both pillars.

Adequacy: Long-Term Planning and Load Forecasting

Forecasting subregional electricity demand over multiyear horizons is complicated by the need to capture and differentiate large loads from general demand growth, introducing substantial uncertainty into planning. AI data center projects often enter transmission interconnection queues in large numbers, with new proposals appearing faster than infrastructure can be planned or built. Due to the low barrier to entry, combined with the speculative and highly competitive nature of access to transmission service, transmission providers and market operators struggle to distinguish viable projects from placeholders. These uncertainties flow directly into long-term planning.

Integrated planning is constrained by a lead-time mismatch where data centers can materialize within months, whereas expanding transmission and generation capacity can take years or even decades, creating a persistent gap in local adequacy. This creates uncertainty in capacity expansion planning, necessitating repeated upward revisions in utility load forecasts. The result is a planning environment where generation and transmission adequacy are constantly strained. Transmission systems are particularly exposed as a single facility or cluster can add hundreds of megawatts of demand at one interconnection point, overwhelming local infrastructure. Traditional assumptions about diversified and gradually increasing loads are no longer viable in regions targeted for hyperscale AI deployment.

Security: Operations Planning and Real-Time Operations

AI workloads exhibit sharp ramping behavior, batch-driven variability, and limited responsive-

ness to market signals. These characteristics complicate short-term demand forecasting and impair the grid's ability to maintain balancing reserves. AI data centers are often price insensitive and inflexible, operating around the clock to meet performance targets. This limits their participation in traditional demand-response programs and creates challenges for grid operators tasked with maintaining real-time equilibrium. Moreover, high uptime requirements reduce operational flexibility and compress the limited windows when grid maintenance can be safely scheduled. Combined with limited visibility into operational profiles, this hinders outage coordination and planned maintenance.

Adequacy and Security: Grid Stability, Power Quality, and Resonance Concerns

The concentration of data centers at single interconnection points or within local pockets creates a new reliability concern: the potential for sudden large-scale loss of load if one or more facilities disconnect unexpectedly. Dense demands within these pockets may challenge voltage stability during severe contingencies, impose network constraints driven by stability limits, and create common-mode failure events. These risks are no longer hypothetical—clusters of nearly 60 data centers totaling more than 1,000 MW have disconnected simultaneously in Northern Virginia on more than one occasion, creating reliability incidents that could have triggered widespread outages. Without fault ride-through capabilities, ramp rate limits, and oscillation protections at data centers, the grid faces growing stability risks. If unaddressed, such risks could propagate systemically, potentially resulting in widespread reliability impacts that remain latent until triggered by a disturbance.

In summary, the reliability challenges of AI data centers stem from the unprecedented scale of interconnection combined with the unique operational characteristics of these loads. These challenges not only span the entire spectrum of grid planning, operations, engineering, and restoration but also extend to policy and regulatory oversight.

Mitigating Grid Reliability Risks

The challenges introduced by large AI data centers have prompted early regulatory responses in several jurisdictions. Policies such as Texas SB6 reflect a growing recognition that traditional planning and interconnection frameworks are not well suited to manage the speed, scale, and operational complexity of emerging large loads.

The electricity sector is addressing these risks through enforceable planning, interconnection, and operating measures to reposition these facilities from static and inflexible loads to controllable and grid-supportive assets.

Interconnection Process Reforms

The rapid growth of large electricity loads has prompted federal and regional efforts to reform interconnection processes that were historically designed for smaller loads. In response, the Secretary of Energy formally asked the Federal Energy Regulatory Commission (FERC) to establish a new standardized process for connecting large load customers (i.e., greater than 20 MW) to the transmission grid. The aim is to make interconnections faster and more predictable by studying large loads together with on-site generation, requiring upfront financial deposits and readiness milestones to ensure project viability as well as offering expedited paths when customers can be curtailed or dispatched as needed. It also clarifies that projects prompting transmission upgrades will cover the costs and gives large customers the “option to build” those upgrades under standard utility oversight. This federal framing complements regional initiatives and is intended to create a predictable national pathway for large customers.

At the regional level, transmission providers are also rethinking standard interconnection processes to better accommodate large, high-impact customers. For example, the Southwest Power Pool (SPP) is working on a new accelerated interconnection process for large loads and

accompanying generation resources, built on several mechanisms: 1) the ability to bring generation to offset grid impacts, 2) flexible load management that allows curtailment, and 3) a shorter interim option for faster connection, in conjunction with an option to acquire firm rights in the longer term. The key elements of this evolving process are summarized in “Key Elements of the SPP Accelerated Interconnection Process for Large Loads.”

These types of new approaches are likely to establish a precedent for other markets and transmission providers to enable fast-tracked interconnection for large customers as they seek to align with federal regulatory actions. Ensuring that new loads and paired generation assets have a clear pathway with regulatory certainty and efficient processes is a key focus for the electricity sector in the years ahead. Regulatory bodies must remain cognizant of the significant impact that these loads can have on BPS reliability. Thus, load-specific requirements, ramp rate limits, metering equipment, power quality, ride-through performance, oscillation mitigation, and voltage and frequency support requirements are critical to harmonize nationally.

Financial Safeguards and Cost Allocation for Reliability Upgrades

Cost allocation is in the spotlight nationally to ensure regulatory gaps do not result in unfair allocation of network upgrade costs to existing ratepayers. Without clear rules, network upgrade costs associated with transmission infrastructure buildout to predominantly serve large load

Key Elements of the SPP Accelerated Interconnection Process for Large Loads

- *Verified requests:* This requires state oversight on load or generation interconnection, including a state certification process to ensure only credible projects submit requests and provide high study deposits to show financial readiness.
- *Certified high-load interconnection level studies:* This establishes standardized study processes to certify whether a load can connect and under what conditions, allowing loads to bring generation to offset transmission congestion and generation deficiency impacts.
- *High-impact large load generation assessment:* This provides a parallel process for evaluating projects that include on-site generation or paired generation support in a coordinated way.
- *Conditional access:* This allows customers to energize facilities sooner, though subject to curtailment if the grid is under stress.
- *Integrated coordination:* This involves bundling design, registration, operations, and technical studies into a single coordinated pathway to interconnection, resulting in efficient system planning instead of siloed planning studies.

At the regional level, transmission providers are also rethinking standard interconnection processes to better accommodate large, high-impact customers.

customers may be shifted to broader ratepayers. Grid operator market monitors have raised concerns about double-digit bill increases, predominantly driven by upgrade costs to serve data center customers. Revised policies should define payment obligations, default penalties, and financial commitment thresholds that reflect the actual impact and uncertainty of large load additions. Provisions that reward demand flexibility, such as prioritized queue placement or interconnection cost discounts, can further align incentives while supporting grid reliability.

Additional safeguards are also being considered and deployed to protect existing ratepayers to ensure fair costs and electricity affordability. Examples include customer-paid interconnection and grid upgrade costs, financial guarantees, contractual commitments and exit penalties, minimum demand charges, demand ratchets, minimum load factor obligations, and other protections. In parallel, new siting obligations could encourage load distribution and reduce bottlenecks. Early disclosure of data center siting plans and regional capacity thresholds could identify early discrepancies between demand and local hosting capacity.

Planning and Forecasting Enhancements

Load forecast uncertainty is at least partly addressed by raising the barrier to entry across all fronts for the load interconnection queue. Interconnection studies should proceed only for projects that pass queue-maturity screening, such as demonstrating site control, financial security, technical completeness, accurate models, etc., thereby reducing placeholders and providing decision-grade data for planning studies. New load is energized in phases under flexible agreements explicitly tied to deliverability milestones, so additions occur only as N-1/N-1-1 capability becomes available. Conversely, colocated or other “bring your own generation” constructs may also enable faster interconnection while also providing energy assets to support adequacy. Forecast fidelity is improved with validated models and standardized load ramp profiles. Grid-enhancing technologies such as dynamic line ratings

may offer some operational network congestion relief while long-lead reinforcements are built. Large AI data centers should also be incorporated directly into reliability and resource adequacy planning, providing ongoing load forecasts, availability profiles, and contingency characteristics into planning models; participating in new ancillary services markets structured for non-traditional assets; and, under defined operating frameworks, integrating into demand-response and system restoration efforts so that they function as reliability assets rather than stressors. Together, these measures can improve visibility into future load growth, reduce planning uncertainty, and align infrastructure timelines with the actual emergence of demand.

Operational Controls and Load Behavior Management

Real-time impacts are reduced by enforcing site-level caps and ramp rate limits on large load customers, which can be accomplished through workload scheduling and supplemental devices such as BES systems (BESSs). Mandating disturbance ride-through will result in more robust installations, settings, and commissioning of grid-interfacing equipment such as UPSs and server-level power conversion settings. Load flexibility, particularly for large load customers with backup or colocated resources that may not be fully utilized, can also drastically improve resource adequacy conditions, particularly during extreme events and energy shortfalls. Small-scale remedial action schemes (RASs) may also be effective to avoid post-contingency network overloads that may occur within local large load pockets. Demand-side and grid-side solutions may be able to turn inflexible loads into flexible assets from an operational perspective.

Grid Interface and Stability Support

Advanced grid-stabilizing solutions are actively being deployed to handle the variability and oscillatory behavior of AI workloads. While solutions are being explored at the chip and rack levels, grid-side solutions may be required to create adequate layers of protection for grid stability

Early disclosure of data center siting plans and regional capacity thresholds could identify early discrepancies between demand and local hosting capacity.

and equipment integrity. For example, enhanced static synchronous compensators (E-STATCOMs) and grid-forming BESSs are being explored for grid-connected and islanded modes of operation due to their fast response times and ability to stabilize load spikes across a wide range of network conditions. Colocated BESSs may also be able to provide fast frequency response and dynamic reactive power support and handle fast ramping and variability—introducing significant opportunities for grid flexibility and reliability enhancements while supporting the energy needs of the data center customers. Regardless, with the advent of complex systems consisting of layered controls and inverter-based technologies, advanced electromagnetic transient studies may be required to ensure reliability for both the grid and the load customer.

Monitoring, Compliance, and Market Integration

Visibility and compliance are supported by high-rate telemetry that enables real-time monitoring, short-term forecasting, and enforcement of ramp rate and availability limits. Instrumentation at the point of interconnection, including phasor measurement units (PMUs), disturbance recorders, and power quality monitors, provides time-synchronized data to verify ride-through performance, recovery behavior, harmonic compliance, and model accuracy. Compliance is maintained through defined test procedures and periodic audits linked to interconnection requirements, with corrective actions triggered by measured deviations. Market mechanisms support reliability by enabling dispatchable load reductions from noncritical AI workloads and compensating fast frequency support from UPS and battery systems through performance-based settlement. These measures collectively support proactive reliability management, verifiable operational performance, and grid-compatible behavior from AI data centers. [Table 2](#) maps the mitigation techniques discussed in this section to the specific reliability challenges introduced by the rapid expansion of large AI data centers.

With reliability risks properly managed, large AI data centers can serve as valuable grid assets

that enhance overall reliability and stability. Leveraging their inherent infrastructure—such as UPS systems and colocated BESSs—enables them to deliver fast frequency response, reactive power support, and other ancillary services. Advanced inverter-based solutions can further strengthen small-signal stability, voltage regulation, weak grid stability challenges, ramp rate smoothing, etc. These benefits also reduce the reliance on conventional reserves. In addition, the computational flexibility of AI data centers enables both temporal and spatial load shifting, improving key grid performance metrics.

Turning Challenges Into Strengths: AI Data Centers as Grid Assets

Power system resilience focuses on the grid's capability to withstand and reduce the magnitude and duration of high-impact, low-probability events (e.g., extreme weather, cyberattacks, and large-scale equipment failures) by anticipating, absorbing, adapting to, and rapidly recovering from such events. In the context of AI infrastructure, resilience frameworks should consider both direct grid vulnerabilities and cascading impacts on other critical services, spanning operational resilience (e.g., updated protection philosophies, RAS designs, and restoration procedures) and infrastructural resilience (e.g., grid hardening, transmission expansion, and revised planning/design standards for large loads).

Resilience in power systems is not a fixed attribute but a dynamic process that evolves through continuous cycles of planning, operations, and post-event learning (see [Figure 4](#)). This framework is increasingly critical for managing new challenges and uncertainties introduced by large AI data centers, and it involves

- 1) *anticipation*: proactively planning for disruptions by integrating AI load characteristics into system design and risk analysis
- 2) *response*: quickly and effectively assessing events, deploying emergency controls, and coordinating restoration
- 3) *learning*: translating post-event findings into refined models, standards, and playbooks that inform future planning.

table 2 Potential solutions to AI data center reliability challenges.									
Solutions	Load Forecasting Uncertainty	Resource Adequacy Misalignment	Transmission Constraints	Real-Time Control Limitations	Grid Stability Issues	Voltage Sensitivity/ Ride-Through Deficiency	Harmonic Distortion/ Resonance	Protection Coordination	Market Inflexibility/ Demand-Response Limitations
UPS				✓	✓	✓		✓	✓
BESS		✓	✓	✓	✓	✓			✓
STATCOM			✓		✓	✓			
Dynamic line ratings and topology optimization		✓	✓						
Workload scheduling and voltage/frequency scaling				✓	✓				
PMUs and monitoring tools							✓	✓	
Staged interconnection and queue maturity screening	✓	✓	✓						
Harmonic filtering					✓	✓	✓		
Market mechanisms (dispatchable load, performance-based settlements)				✓	✓				✓

Data centers can contribute to grid resilience in meaningful ways, provided that policy and regulatory frameworks evolve to enable such opportunities.

Framing resilience as this closed loop builds systematic learning into the grid's architecture, ensuring that as large AI data centers reshape the load landscape, resilience evolves in parallel, guided by real-world system behavior and emerging threats.

Unlocking the Resilience Potential of AI Data Centers

This section pivots away from the familiar narrative that casts AI data centers as “power-hungry” sinks with predominantly negative grid impacts. Instead, it envisions a near-term future where roadblocks are overcome and these facilities operate as grid-supportive assets. This outlook envisions that innovation will solve many of the grid reliability risks through cross-sector collabora-

tion and that the data center industry will act as a good steward of grid reliability. Thus, creative pathways to integrate AI data center loads at scale can be pursued by deploying technologies and strategies that make their demand grid-compatible and supportive. Looking ahead, data centers can contribute to grid resilience in meaningful ways, provided that policy and regulatory frameworks evolve to enable such opportunities.

Temporal Flexibility for Peak Relief and Variability Reduction

Data centers may provide significant levels of flexibility to the grid if they are able to ramp their workloads and power demands to meet grid operator needs. This may include curtailing or reducing power demand during resource adequacy or energy adequacy shortfalls, extreme events, and other emergency operating conditions [see Figure 5(a)]. Having spatially concentrated “mega loads” across the grid that demand firm service may create challenges for grid operation and reliability and require significant infrastructure investment. Turning these loads into flexible and curtailable assets for grid operators may result in a future grid that is able to handle higher levels of variable and intermittent resources moving forward.

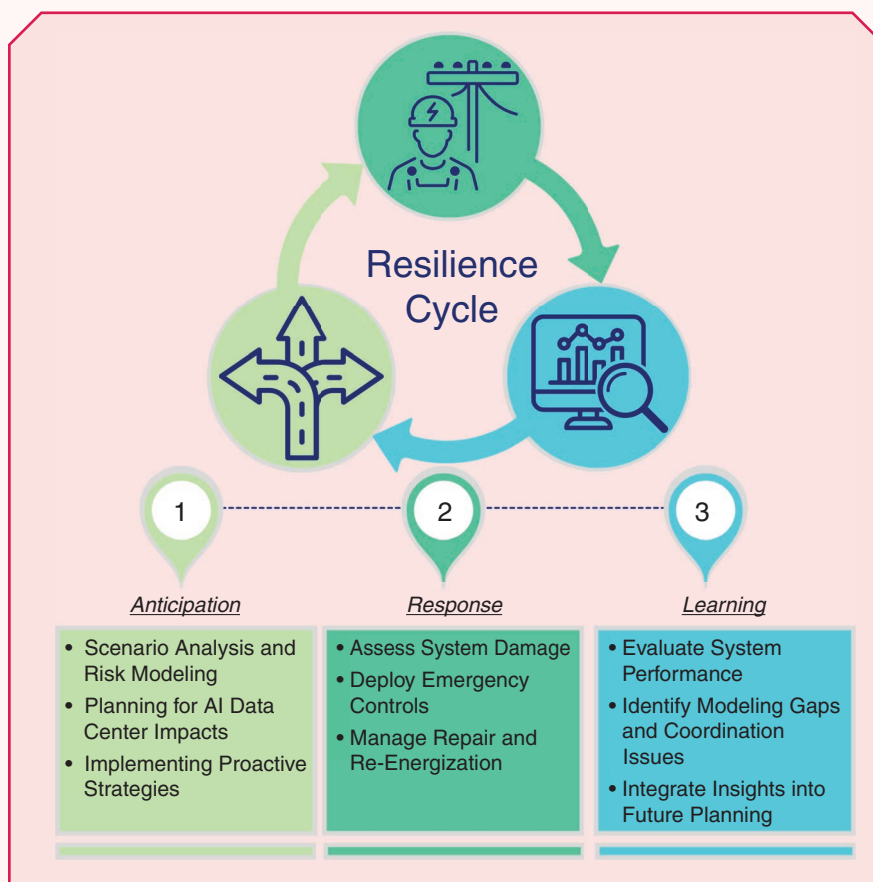


figure 4. The resilience cycle under AI data center growth.

Spatial Flexibility for Regional Stress Mitigation During Extreme Events

Extreme events are also generally regionally concentrated, and data centers may play a key role in shifting workloads to other synchronized data centers in other parts of the

grid or even in different interconnections [see Figure 5(b)]. In practice, spatial flexibility is primarily applicable to inference and batch-processing workloads, which can be shifted across geographically distributed data centers without violating latency or data-locality constraints. In contrast, large-scale training remains geographically constrained due to data gravity, high bandwidth demands, and the need for tightly coupled high-density GPU clusters. The ability to offload regional grid stresses during wildfires, hurricanes, derechos, and other extreme hot and cold conditions could provide grid operators with additional tools to manage emergency conditions.

Leveraging On-Site Resources for Faster Restoration

Data centers are deployed with a tremendous amount of on-site backup generation. These generators often sit dormant until a grid event happens and the UPSs initiate a transfer over to these on-site generators. Historically, these generators have been diesel gensets that are limited to a small number of operating hours due to environmental restrictions. However, various colocated and backup generation solutions, including natural gas, renewables, and BESSs, are emerging. Under extreme events or during system restoration efforts, leveraging the full capability and capacity of these resources could enable quicker restoration and higher reliability. However, policies need to evolve to consider these options.

Enabling Resilience Through Data Center Performance Standards

There are currently few technical requirements—capability, performance, and utilization—that define how large loads such as data centers must operate when connected to the grid. While power quality and interconnecting protection system requirements may exist, broader requirements related to ride-through performance, ramp

rates, variability limits, oscillation mitigation, etc., are not well established in the United States and present significant uncertainties to grid planning, design, and operations. Shoring up these gaps could further result in data centers being a reliable and trustworthy load that supports broader resilience efforts.

Dual Role of AI Data Centers in Grid Support

Fully leveraging the unique operational capabilities of data centers and maximizing the value that they can bring to grid operators can improve local, regional, and BPS-wide stability, reliability, and resilience moving forward. Operationally, we may be able to think about data centers as hybrid assets, simultaneously serving as a highly flexible load that additionally includes dispatchable resources that support energy policy goals. This can enable improved grid support services such as ramping, peak shaving, improved fault ride-through and stability, quicker system restoration,

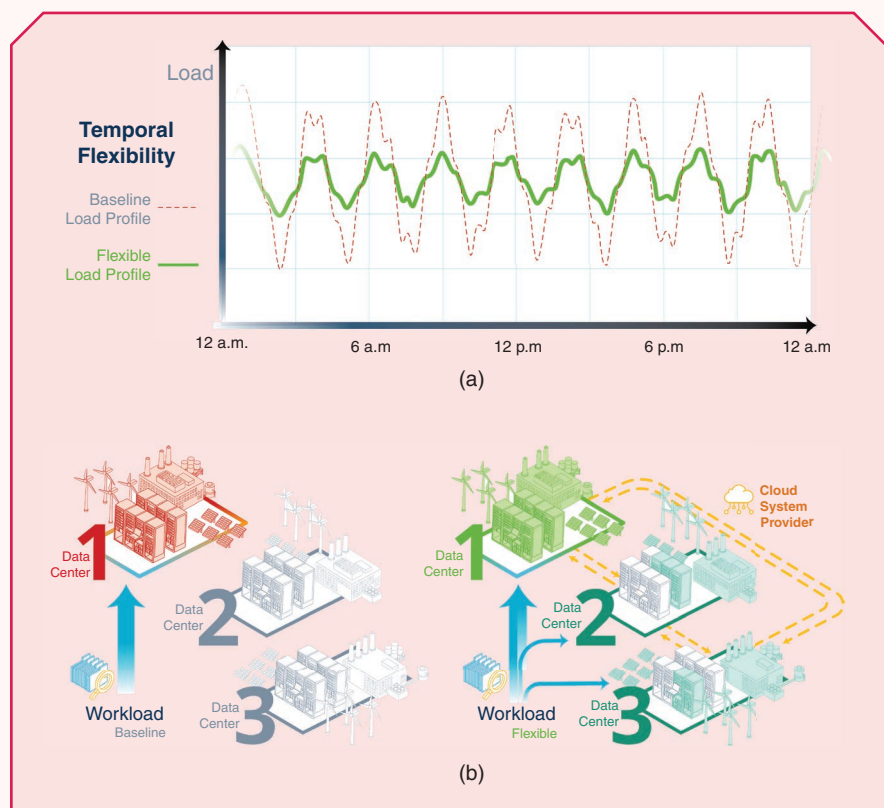


figure 5. (a) A temporal flexibility pathway for AI data centers. Temporal flexibility is achieved by modulating or shifting compute over time, ramping workloads and power demand to align with grid conditions, to flatten peaks, fill valleys, and reduce intraday load variability. (b) A spatial flexibility pathway for AI data centers. Spatial flexibility is achieved by dynamically routing workloads across geographically dispersed facilities, optimizing resource utilization, and mitigating local grid stress.

resource adequacy deficiencies, and other resilience benefits.

Realizing the resilience value of AI data centers requires clear operational roles and compatible interconnection arrangements for hybrid facilities that are both large loads and holders of dispatchable storage/generation. Streamlined processes that recognize this dual nature (e.g., coordinated treatment of load and generation interconnections, defined emergency operating modes, and explicit compensation mechanisms for event-time services) reduce procedural friction and align incentives. Policy adjustments should focus on specifying when and how these facilities support the grid during emergencies, defining data and control interfaces to enable safe coordination, and ensuring that cybersecurity and reliability requirements are met for any grid-support function.

Looking Forward

The rise of AI and hyperscale data centers is reshaping electricity demand in ways that have profound impacts on grid reliability and resilience. Meeting these demands requires more than steel in the ground and wires in the air. Looking ahead, the electricity sector needs to remain focused on innovations in the following areas:

- improving power system modeling practices for large loads across multiple simulation timescales and domains
- incorporating the performance characteristics of large loads into studies for grid interconnection, long-term planning, operational planning, and real-time applications
- establishing streamlined and equitable interconnection processes that facilitate the connection of both new large loads and resources to meet energy and capacity demands
- ensuring clear, transparent, and fair cost allocation that protects existing ratepayers while facilitating economic growth and development
- advancing policies and operating practices that fully leverage colocated generation and loads to maximize grid flexibility, particularly during generation shortfalls or grid congestion
- fostering market-driven innovation to bring value to customers and enable novel approaches to connect large loads effectively and economically.

Achieving such a vast undertaking will require sustained investment in workforce development. Building and maintaining the grid of

the future will depend not only on technological advances but on the technicians, engineers, planners, operators, and skilled tradespeople who make those advances real. Anticipating the pace of technological change and making strategic investments in research and development are equally critical, ensuring that new technologies are practical, implementable, and ready to meet evolving grid demands. Embedding resilience into every layer of planning and operations and equipping the power sector to adapt are key to keeping the lights on and the information flowing. If utilities, regulators, and innovators stay focused on these priorities, the grid will remain a reliable, resilient, and economic foundation upon which modern society is built.

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For Further Reading

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